MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, SILICON, LOW-POWER
TYPES 2N328A, 2N329A, 2N329AS, and 2N329AS

This amendment forms a part of MIL-S-19500/110C, dated 11 April 1972, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title, delete and substitute new title as printed above.

1.1, add the following:

"The suffix 'S' is used on devices that have 0.5 inch minimum to 0.75 inch maximum lead length."

1.3, add the following new type numbers:

Following "2N328A" add "2N328AS"; following "2N329A" add "2N329AS". Make these changes wherever they occur throughout the document.

PAGE 2

Add the following new paragraph:

"3.5.1 'S' marking. The 'S' suffix shall be used on devices meeting the 0.5 inch minimum to 0.75 inch maximum lead length requirement."

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
3	3
4	4

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - MI
Air Force - 11, 80
DLA - ES

User activities:
Army - AV, SM
Navy - AS, CG, MC, OS, SH
Air Force - 13, 15, 19

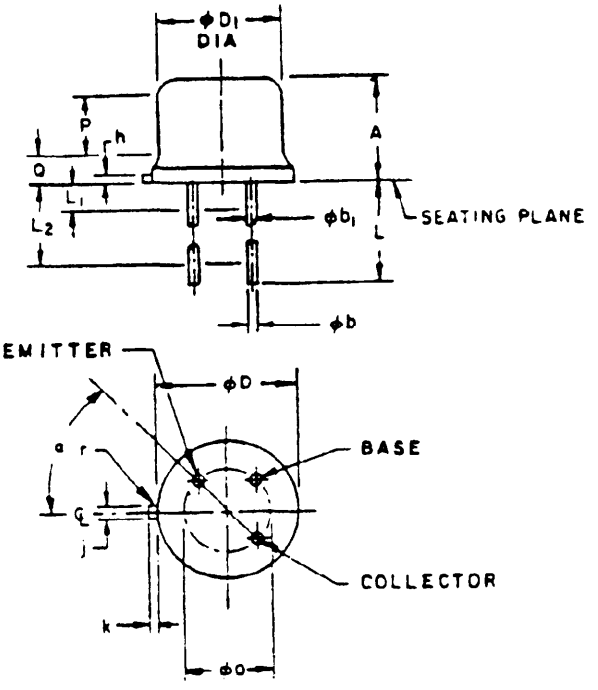
Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5961-1019-2)

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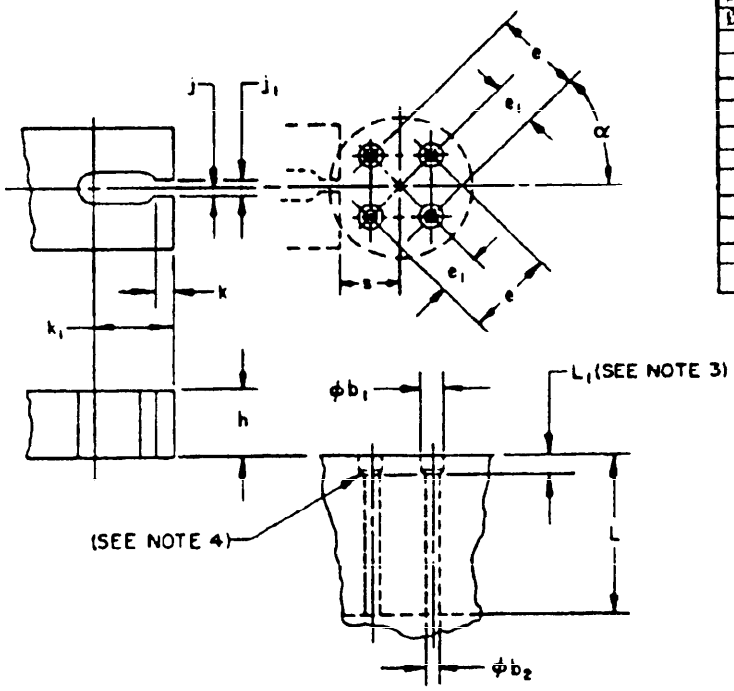


Ltr	Dimensions				NOTES
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.240	.260	6.10	6.60	
Øa	.200 TP		5.08 TP		6
Øb	.016	.021	0.14	0.53	7, 8
$Øb_1$	.016	.019	0.41	0.48	7, 8
ØD	.335	.370	8.51	9.40	
$ØD_1$	.305	.335	7.75	8.51	
h	.009	.041	0.23	1.04	
j	.028	.034	0.71	0.86	2
k	.029	.045	0.74	1.14	3
L	See notes 7, 8, 11, and 12				
L_1	---	.050	---	1.27	7, 8
L_2	.250	---	6.35	---	7, 8
P	.100	---	2.54	---	5
Q	---	.050	---	1.27	4
r	---	.010	---	0.25	10
α	44° TP		44° TP		6

NOTES:
1. Metric equivalents are given for general information only.
2. Beyond r(radius) maximum, j shall be held for a minimum length of .011 (0.28 mm).
3. k measured from maximum ØD.
4. Outline in this zone is not controlled.
5. $ØD_1$ shall not vary more than .010 (0.25 mm) in zone P. This zone is controlled for automatic handling.
6. Leads at gage plane .054 + .001 -.000 (1.37 +.03 -.00 mm) below seating plane shall be within .007 (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by the gauge and gauging procedure shown in figure 2.
7. $Øb_1$ applies between L_1 and L_2. Øb applies between L_2 and L minimum. Diameter is uncontrolled in L_1 and beyond L minimum.
8. All three leads.
9. All leads electrically insulated from the case.
10. r(radius) applies to both inside corners of tab.
11. For transistor types 2N328AS are 2N329AS, L is .500 (12.70 mm) minimum, and .750 (19.05 mm) maximum.
12. For transistor types 2N328A and 2N329A, L is 1.500 (38.10 mm) minimum, and 1.750 (44.45 mm) maximum.

FIGURE 1. Physical dimensions.

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LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
ϕb_1	.0595	.0605	1.51	1.54	
ϕb_2	.0325	.0335	.83	.85	
e	.1995	.2005	5.07	5.09	
e_1	.0995	.1005	2.53	2.55	
h	.150 Nominal		3.81 Nominal		
j	.0175	.0180	.44	.46	
j_1	.0350	.0355	.89	.90	
k	.009	.011	.23	.28	
k_1	.125 Nominal		3.18 Nominal		
L	.372	.378	9.45	9.60	
L_1	.054	.055	1.37	1.40	
S	.182	.199	4.62	5.05	1
α	44.90°	45.10°	44.90°	45.10°	

NOTES:
1. The location of the tab locator within the limits indicated will be determined by the tab and flange dimensions of the device being checked.
2. The following gaging procedure shall be used:

   The device being measured shall be inserted until its seating plane is .125(3.18 mm) ±.010(.25 mm) from the seating surface of the gage. A force of 8 ± .5 oz. shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gage.

   The use of a pin straightener prior to insertion in the gage is permissible.

3. Gaging plane.
4. Drill angle.

FIGURE 2. Gage for lead and tab location.

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